

CARBON REMOVALS EXPERT GROUP TECHNICAL ASSISTANCE

Technical assessment of certification methodologies for long-term biogenic carbon storage in buildings

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1. Summary of main findings

Drawing upon the insights and inquiries highlighted within this report, the following set of recommendations has been outlined to inform and guide future assessments. These recommendations cover key considerations and areas necessitating additional exploration to advance understanding and practice.

Quantification

Technical recommendations

1. **Exclusion of end-of-life stages:** End-of-life stages (EoL) and beyond should not be considered within the scope due to uncertainties in predicted EoL and data availability. Circularity practices, such as reuse and recycling, can be incorporated either into minimal sustainability criteria or incentivised through the co-benefits mechanism. However, further consideration should be given to their future inclusion when these practices become more prevalent and data availability improves.
2. **Standardised baseline:** The use of a standardised baseline is advised to maintain a consistent, comparable, and transparent calculation of the baseline. A recent study into baseline values for embodied and stored carbon in buildings in the Union can be used to establish the figures. A standardised CR_{baseline} needs to be calculated based on baseline figures for carbon storage per square meter in typical buildings, expressed in $\text{kg CO}_2/\text{m}^2$. These figures must be differentiated across various building types and geographic locations.
3. **Periodic re-evaluation of market share:** The market share of biobased construction elements should be periodically re-evaluated, and the baseline figures for the standardised baseline adjusted accordingly. In the beginning, using projections of an evolving baseline over time is not advisable.
4. **Activity-specific baseline:** It is recommended to utilise a standardised baseline rather than an activity-specific one in this certification scheme, as proposed above. However, should the decision be made to switch to an activity-specific baseline in later stages, the operator must establish a baseline storage factor with identical units ($\text{kg CO}_2/\text{m}^2$) sourced from publicly available and verifiable data. According to CRCF Regulation, if an activity-specific baseline is chosen, additionality tests become necessary, and it is advisable to incorporate a discount factor for uncertainty.
5. **Calculation of CR_{total} :** CR_{total} shall be calculated by summing the biogenic carbon content of biobased construction elements in the project. The biogenic carbon content for each eligible construction element can be derived from environmental product declarations (EPDs). Specifically, the GWP-biogenic indicator for stages A1-A3, as stipulated in EPDs adhering to the EN15804+A2 standard, can be utilised. In cases where EPDs are unavailable, alternative reliable sources may be considered, pending further deliberation on their acceptance and prioritisation.
6. **Calculation of $GHG_{\text{associated}}$:** The method should prescribe a standardised GHG_{baseline} expressed in $\text{kg CO}_2 \text{ eq.}/\text{m}^2$, differentiated for regions and building types. GHG_{total} emissions for the target building are determined as the sum of GWP-fossil and GWP-

luluc indicators in stages A1-A5 for all construction elements. The $GHG_{baseline}$ multiplied by the floor surface area is subtracted from the GHG_{total} to compute $GHG_{associated}$.

7. **Applicability of certification methodology:** The certification methodology can be used for carbon storage in both new buildings and renovation projects, provided reference values for $CR_{baseline}$ are determined and provided for renovated and retrofitted buildings.

Further deliberation required

8. The scope of the calculations for $CR_{baseline}$, CR_{total} , and $GHG_{associated}$ should encompass the production and construction stages from raw material supply to the manufacturing of construction elements, including transportation steps (A1-A5), to adhere to the existing standard (EN15804+A2) to the extent possible. However, should the GHG emissions in the A1 stage be included in the calculation of $GHG_{associated}$? If the certification is only applicable to construction elements produced from biomass that adheres to the criteria set out in Article 29 of the Renewable Energy Directive (RED), the A1 emissions might be excluded to be in line with Article 29 of the RED.
9. Standard emission factors for indirect land-use change must be established for common biobased construction materials, building on indirect land use change (ILUC) emission factors in RED, where applicable.
10. Further exploration is needed to consider various options for calculating $GHG_{associated}$, aiming to reduce administrative burden for all involved parties.
11. Addressing uncertainty in data used for quantifying net carbon removal benefit is crucial. While deducting percentages of certifiable stored carbon appears to be the best practice, determining these percentages warrants further discussion.
12. Cross-cutting issues across certification methodologies for carbon farming and biogenic building materials storage such as preventing double counting and avoiding overestimation of the potential harvested wood products (HWP) pool for national GHG inventories will be addressed in further discussions.

Additionality

Technical recommendations

13. The use of a standardised baseline is advised to maintain a consistent, comparable, and transparent calculation of the baseline. A recent study into baseline values for embodied and stored carbon in buildings in the Union can be used to establish the baseline figures. A methodology is currently in development that can be used to regularly update these baseline figures.
14. If after further deliberation is chosen to use an activity-specific baseline instead of a standardised baseline for the certification methodology, operators must disclose any relevant regulations and financial incentives supporting the utilisation of biobased construction materials in their projects. They must also clarify how the carbon storage

is additional. To assist operators in their own markets in this process, Member States could compile a list containing local regulations and financial mechanisms that encourage the adoption of biobased construction materials.

15. If the activity-specific baseline is used, the mandatory use of biobased materials due to regulatory requirements or financial incentives, apart from those associated with the certification methodology, must be disclosed in the certification application. Such usage should be excluded from the calculation of CR_{total} .

Further deliberation required

16. Options to prove and verify financial additionality that do not require full project financials should be further explored.

Long-term storage

Technical recommendations

17. The CRCF regulation mandates periodic re-certification audits for all activities at least every five years, deemed adequate for monitoring biobased construction products. Eligibility is restricted to elements with a minimum 35-year lifespan, preferably focusing on structural elements and insulation materials.
18. Utilising a buffer pool as collective insurance against unplanned releases is recognised as the best available practice in current methodologies. Determining the portion of total certifiable carbon storage allocated to the buffer pool remains undecided. National insurance data is suggested for assessing region-specific risk of reversal.

Further deliberation required

19. What percentage or range of percentages is optimal for allocating to a buffer pool to address financial liability?
20. Further deliberation is necessary to identify effective risk mitigation mechanisms aimed at minimising the risk of carbon storage reversal.
21. How can insurance agencies contribute to mitigating financial liability in the event of unexpected carbon release?
22. There is a need for continued discussion on how the certification methodology can incentivise building owners to extend the carbon storage lifespan beyond the minimum requirement of 35 years.

Sustainability

Technical recommendations

23. The 'Sustainability and greenhouse gas emissions saving criteria for biofuels, bioliquids and biomass fuels' from the Renewable Energy Directive ([RED III](#)) can be used as a template to establish minimum requirements regarding this sustainability objective. While the direct linkage of minimal sustainability requirements to this regulation faces challenges due to the diverse uses of biomass—ranging from raw material to energy—Article 29 of RED III offers a solid foundation.

Further deliberation required

24. Additional discussions are needed regarding the minimum sustainability requirements, with a focus on aligning these requirements with the DNSH criteria of the taxonomy across all objectives. Specifically, for the first four objectives—climate change mitigation, climate change adaptation, sustainable use and protection of water and marine resources, and transition to a circular economy—consideration should be given to incorporating indicators and assessment methodologies outlined in the Level(s) framework.
25. The upcoming iteration will delve into establishing guidelines for harmonized reporting of sustainability co-benefits, building upon the Taxonomy or other relevant EU policies as deemed appropriate.

2. Introduction

It should be noted that the information contained in this document reflects the best available knowledge as of March 2024. While updated to the [provisional agreement](#) on the establishing a Union certification framework for permanent carbon removals, carbon farming and carbon storage in products (CRCF Regulation) have been incorporated, some adjustments have prompted the need for further deliberation in subsequent phases of the assessment.

2.1 Goal

This document serves to inform the carbon removals expert group on the on the advancements in developing technical advice for certifying long-term biogenic carbon storage in buildings, according to the provisional agreement on the CRCF Regulation. It compiles insights from previous discussion, expert inputs, and analyses of existing methodologies. CRETA's consultants facilitated focus groups, engaging a diverse set of experts, though these were independent of CLIMA and may not entirely represent the collective opinion of the entire Expert Group. The purpose of this report is to serve as a guide for further discussions on the topic and address open questions.

Each chapter focuses on a quality criteria, outlining and discussing the options to cover the provisions of the CRCF Regulation, noting the best available option while also highlighting questions arising during the review of potential methodologies for application. Many sub-criteria are interconnected across the four quality criteria. Proposed solutions may draw support from other criteria, or issues may already be addressed under different criteria. Conclusions and unresolved questions often reference other sections within the document.

The outcome of this process presents three key points for orientation:

1. **Best available practice:** A best available practice in existing methodologies is provided and explained in further detail. It is crucial to acknowledge that, while these practices represent the current best practice, there exists potential for further development and improvement. Instances where refinement is possible are outlined with accompanying explanations.
2. **Ongoing discussion:** This captures the ongoing discourse among experts regarding what constitutes the best practice. Conclusions from the CRETA consultants based on the discussions and review of existing methodologies are presented.
3. **Open questions:** If little or no discussion has taken place, or confusion and a lack of clarity still exists on a particular aspect, this is noted as an open question. Suggestions for potential solutions or comments that were made on the subject are noted. These sections serve as a placeholder for areas where further exploration, research or consensus may be required.

2.2 Context

The provisional agreement on the CRCF Regulation aims to contribute to the EU's climate and environmental goals and aligns with other EU initiatives. In December 2021, the Commission addressed the importance of developing a standard, robust, and transparent methodology for quantifying the potential carbon storage benefits of construction products in the [Sustainable Carbon Cycles Communication](#).

This commitment is reinforced by various Commission initiatives, such as the [2030 Forest Strategy](#), the revised [Construction Product Regulation](#), the revised [Energy Performance in Buildings Directive](#), the revised [LULUCF Regulation](#), and the [New European Bauhaus](#). These initiatives all refer to promoting long-lasting carbon storage in construction products.

Simultaneously, stakeholder meetings on buildings and construction products consistently express interest in quantification of carbon storage in construction products, in particular in the preparatory work for the development of harmonised standards under the Construction Products Regulation. Several Member States are already (working on) incorporating carbon storage in building codes. This collective effort aligns with the EU's vision for a more sustainable and low carbon construction industry.

2.3 Overview: Concept direction for certification

The European Parliament will vote on the CRCF Regulation in April 2024. Once adopted, the legal text will enter into force upon its publication in the EU Official Journal, tentatively by the end of 2024. We then move on to implementation, including the adoption of EU certification methodologies, third-party verification rules, EU recognition of certification schemes and set-up of EU-wide registry.

Based on the analysis presented in this report, the following conceptual trajectory emerges:

The certification will primarily target biogenic carbon stored within structural (load bearing) elements and potentially biobased insulation materials of buildings, aiming to mitigate the risk of unintended reversal and optimise storage potential. Both new construction and renovation projects will qualify for certification, with products expected to have a minimum lifespan of 35 years.

The CRCF Regulation differentiates between permanent and temporary removal units. This means that temporary units from carbon storage in products expire at end of the relevant monitoring period, unless the monitoring period is renewed, or permanent storage is demonstrated by the operator.

The certification methodology may evolve over time through delegated acts, with re-certification mandated at least every five years. Integration of certificate verification and monitoring into existing building check-up routines will be pursued.

To prevent unwanted burden shifting and rebound effects, minimum sustainability requirements will be established for projects, materials, and material sources. Further details on these requirements are available in the sustainability chapter.

Building owners, as the intended recipients of the certified units, can utilise the units in various ways. They can be traded on the voluntary carbon market, utilised to declare the carbon storage indicator in Energy Performance Certificates (based on Article 16 and Annex V of the [Energy Performance of Buildings Directive \(EPBD\)](#)), or employed to substantiate claims regarding carbon storage in alignment with the [Corporate Sustainability Reporting Directive \(CSRD\)](#).

With the revised EPBD allowing building owners to declare the carbon storage capacity of their structures on their Energy Performance Certificate (EPC), they can provide reliable evidence through the CRCF certification methodology for carbon storage in products and related carbon storage in product units. This enables them to demonstrate their buildings' carbon storage capacity transparently. For example, construction companies or property owners investing in long-term sustainable building materials, such as wood-based ones, could earn additional income through the sale of CRCF units.

Carbon removals and carbon storage in the EPBD recast

In alignment with the recast directive, specifically Article 7, Member States are directed to address various aspects concerning new buildings, including carbon removals associated with carbon storage in or on buildings. Moreover, Annex II of the directive outlines the template for national building renovation plans, referring to Article 3, wherein Member States are required to provide an overview of implemented and planned policies and measures, including the reduction of whole life-cycle greenhouse gas emissions for construction, renovation, operation, and end-of-life phases of buildings, as well as the uptake of carbon removals. Additionally, Annex V details the template for energy performance certificates, as referred to in Article 16, which may include information on carbon removals associated with the temporary storage of carbon in or on buildings.

3. Quantification

3.1 Provisional agreement on CRCF Regulation

This criterium refers to Article 4 of the CRCF regulation. Essentially, a carbon removal activity shall be quantified in a relevant, conservative, accurate, complete, consistent, comparable and transparent manner, in accordance with the latest available scientific evidence. The temporary net carbon removal benefit shall be quantified using the following formula:

$$\text{Temporary net carbon removal benefit} = CR_{\text{baseline}} - CR_{\text{total}} - GHG_{\text{associated}} > 0$$

- (a) CR_{baseline} is the carbon removed under the baseline;
- (b) CR_{total} is the total carbon removals of the carbon storage in products activity;
- (c) $GHG_{\text{associated}}$ is the increase in direct and indirect greenhouse gas emissions, over the entire lifecycle of the activity which are due to its implementation, including indirect land use change, calculated, where applicable, in accordance with protocols set forth in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and any further refinement.

The baselines shall be highly representative of the standard performance of comparable practices and processes in similar social, economic, environmental, technological and regulatory circumstances and take into account the geographical context including regulatory conditions ('standardised baselines'). Where duly justified, an operator shall use a baseline that corresponds to the individual, performance of a specific activity ('activity-specific baseline'). In this case, additionality shall be demonstrated through specific tests. Furthermore, the quantification of the carbon removals shall account for uncertainties in a conservative manner and in accordance with recognised statistical approaches.

Aspect	Status
Scope of calculations	Ongoing discussion
Quantification of a standardised CR_{baseline}	Ongoing discussion
Quantification of an activity-specific CR_{baseline}	Ongoing discussion
Quantification of CR_{total}	Best available practice
Quantifying $GHG_{\text{associated}}$	Ongoing discussion
Accounting for uncertainties	Ongoing discussion
Including both new buildings and renovations	Best available practice

3.2 Scope of calculations

Ongoing discussion

Definition	The scope refers to the system boundaries for the calculation, which lifecycle stages to include in the calculation and which processes and resources associated emissions are included within these lifecycle stages.
Issue	The challenge is to include the relevant stages of the life cycle of the building or construction product. These stages become particularly relevant when comparing biobased materials to conventional building materials, as various factors such as land use change, transport distances and production process contribute to distinctions. Consequently, a crucial aspect within the certification scheme is determining which parties bear the associated GHG emissions. For instance, in Stage A1, the question arises: should forest owners or sawmills be allocated these emissions? Therefore, within life cycle the establishment of clear distinctions or sub-boundaries may be needed for determining the responsibility for GHG emissions throughout the process. Another consideration emerges when incorporating later stages of the life cycle, such as Phase C in the EN15804 standard. If these stages are included, the overall balance reported for storage becomes zero. In other words, all the carbon sequestered during the growth phase, stored in the products/buildings, is released at end of their life cycle (i.e. the release of the stored carbon is +/-). This results in negating the benefit of the “temporary” carbon storage.
Objectives	Clearly define which stages of the lifecycle of biobased construction products should be included in the calculation of $CR_{baseline}$, CR_{total} , and $GHG_{associated}$.
Existing, proven certification methodologies	The project scope for assessing the embodied and stored carbon varies in existing methodologies. Roughly, two thirds of the reviewed methodologies include the production stages of bio-based products within their scope. Many of these methodologies focus on the product stages A1-A3, with some extending their scope to include the disposal stage C4. Only a few methodologies include the use phase, mainly the general LCA methodologies rather than the certification methods. No methodology considers the circularity-related stage D. Diagram of life cycle stages based on BS EN 15978 / PAS 2080 / EN 15804 and ISO 21930 adapted by Designing Buildings

The following options are not mutually exclusive, a selection must be made of stages to be included in the scope of the calculations.		
Options	Pros	Cons
Include agricultural and forestry practices (A1)	• Biogenic carbon removal from the atmosphere is captured in this stage. If it is not included, biogenic carbon removal as included in EPDs cannot be used to calculate CR_{total}. • Attributable emissions can vary significantly between different biobased resources and land management practices. • Unsustainable forestry and land-use practices would be discouraged by the methodology. • Discourages the exploitation of far-off natural sources with harder-to-verify sustainability certification and emission values. • Closer link with sustainability criteria.	• Might require data inquiry and validation at tier 2 and tier 3 suppliers, increasing effort for data collection and decreasing reliability/comparability. • Might overlap with certification of carbon removal/reduction in forestry and agriculture. • Regarding the scope of $GHG_{associated}$, harvesting emissions for sustainably harvested biomass that adheres to the criteria set out in Article 29 of the Renewable energy directive (RED) might be excluded, to be in line with the RED and the zero-rating of biomass in the ETS¹. • The availability of other approaches that ensure sustainable land-use (e.g. as a requirement for material eligibility or considering its inclusion within a co-benefit mechanism) might make the inclusion of this stage in $GHG_{associated}$ redundant.
Include transport of raw material (A2) and manufacturing of constructing elements (A3)	• Indirectly incentivises local sustainable sourcing of raw materials and/or sustainable transport. • EPD methodology EN15804 can be used to uniformly assess direct and indirect emissions from manufacturing methods. • Ongoing EC study into GHG emissions of EU buildings and construction (Ramboll, 2023²) caution about the potential for significant emissions and environmental burdens associated with glue and paint use in timber construction.	• Data acquisition and inventory development could be challenging for novel biobased products. • Additional boundary consideration may need to be addressed for recycling and refurbishment.
Include installation of the biobased construction elements in the construction stage of the building (A4 & A5)	• Including transport (A4) incentivises local sourcing of products with (assumed) better oversight of working conditions and production emissions and	• The stages A4 and A5 are not mandatory to report in an EPD according to EN15804+A2. The EPDs commonly report on a product-level, general values for

¹ It is important to note that the biomass carbon accounting in RED is based on the UNFCCC inventory principle of counting biogenic CO₂ emissions as zero in industrial inventories because changes in standing biomass carbon stocks are to be dealt with in the land use, land use change and forestry sector.

² Rambol (2023) *Analysis of Life-Cycle Greenhouse Gas Emissions of EU Buildings and Construction*
<https://ec.europa.eu/docsroom/documents/58196>

	improvement of local sustainable economic opportunities. • Fossil-based transport of large construction elements can have an impact, although the comparison with conventional construction elements remains uncertain.	transport to a construction site and construction-associated emissions cannot be included in these reports. • Distinguishing the point at which a structural construction element is produced from the point where the construction stage begins can be challenging, especially with prefabricated modular biobased elements like walls with insulation and window frames. Determining system boundaries may require additional work. • Identifying which construction activities can be attributed to biobased elements versus conventional elements is challenging, complicating the allocation of associated emissions. Whether this is relevant depends on how GHG_{associated} is calculated. Refer to section 2.5 “Quantification of GHG_{associated} for more on this.
Include the use of the building (B1), specifically the effect of the biobased construction element on resource consumption for heating, cooling, and amenities.	• Over the lifespan of a building, small differences in, for instance, insulation properties can lead to significant variations in associated emissions for heating and cooling.	• Inclusion of these aspects in the calculation requires data on the operational performance of the building and assumptions about its use over the lifespan of the building, for both the target building and a representative baseline. This requires extensive effort in data collection and validation. Experts are against including it in the scope. • Other approaches are available to prevent the certification from incentivising unsustainable material choices in these aspects. For instance, certifying only structural construction elements or exclusively certifying buildings with a certain minimal energy performance.
Include the rest of the use stage: maintenance, repair, replacement, refurbishment (B2 – B5)	• Maintenance and repair can lead to significant variations in associated emissions throughout the lifespan of a building. • Actual data on maintenance, repair, and replacement since the initial certification can be integrated in a re-certification process.	• Inclusion of these aspects in the calculation requires assumptions about the lifespan of construction elements and resource consumption in their maintenance for both the target building and a representative baseline. However, these assumptions can be informed by through EU projection studies into the maintenance and repair performed on defined building

		types in various regions in the Union.
Include the end-of-life stage (C1-4)	• Biobased construction elements can undergo different disposal methods, resulting in either a rapid release of stored carbon (through incineration) or a gradual release of carbon over time (via biodegradation, e.g. landfilling). This emission process may generate different GHG emissions (CO, CO₂, CH₄) with different GWPs. Both aspects, release time and GHG type, may have a significant effect on the warming effect in the short term (decades).	• Any stored biogenic carbon will eventually be released at some point at end of life. Including this release in the calculation will per definition lead to no carbon removal benefit to be certified, only a time-limit storage of biogenic carbon. • Predicting the end-of-life for building elements becomes challenging, especially without definitive proof of handling after a lifespan exceeding 35 years.
Include potential recycling or reuse after end of life (D)	• Inclusion of the D stage would incentivise reuse, cascading use, and recycling, promoting a circular economy. • Designing construction elements with recycling or reuse in mind can be incentivized in the first use cycle.	• Assumptions about the fate of materials or products many years in the future need to be made. • Effort would be required to align such an approach with other policies and directives. • Additional administrative effort would be needed to account for or audit the inclusion of such elements. • Design for reuse and recycling can be included and rewarded through the voluntary sustainability co-benefit mechanism. See further deliberations on this under 'Selection of co-benefits' in the Sustainability chapter.
Technical conclusions	1. It is recommended to adhere to the existing standard (EN15804+A2) to the extent possible, emphasising the inclusion of production phase emissions (A1-A5). 2. Experts have flagged the allocation issue regarding construction emissions of a whole building to biobased construction elements. Despite this challenge, the inclusion of the construction process stage (A5) in the assessment is preferred. As the project developer or building owner will likely apply for the certification, they can be held responsible for all emissions up until the building is delivered. This incentivises emission reductions in transport and construction and captures material losses occurring at the construction site. 3. It is recommended to exclude the end-of-life stage (denoted as the "D stage"). Predicting the ultimate fate of building elements at the true end-of-life stage proves exceedingly difficult, especially for products with lifespans surpassing 35 years. Instead, the benefits of reusing construction elements in a second lifecycle or recycling materials after their end-of-life stage should	

	be included in the minimal sustainability criteria or the co-benefit mechanism.
Open questions	• Should GHG emissions from the A1 stage be included in the calculation of GHG_{associated}? If the certification only applies to construction elements produced from biomass meeting the criteria set out in Article 29 of the Renewable Energy Directive (RED), excluding A1 emissions aligns with Article 29 of the RED. • What should be the cut-off point in the A1 stage between carbon farming and carbon storage in products? Initiating the A1 stage at the extraction point of biobased materials, essentially starting from harvesting for GHG-associated purposes, is proposed. While this approach is suitable for timber sourced from certified origins, it may not universally apply to all biobased materials. Excluding agricultural practices and land use change would be inappropriate for crops specifically cultivated for construction materials such as hemp, flax, and miscanthus. Regarding timber, this recommendation holds true only if the eligibility criterion mandates the use of certified sustainably sourced materials from within the EU. • How can the emissions in the A4 and A5 stage be quantified in a consistent and comparable manner, if not referenced from EPDs? • How can the emissions associated with the building construction (A5 stage) be allocated to the biobased construction elements?
Next steps	1. Evaluate whether GHG emissions from the A1 stage should be included in the calculation of GHG-associated emissions. Consider aligning with Article 29 of the RED and existing EN standards, while ensuring alignment with other certification methodologies as well, like the biomass production or collection emission calculations for BECCS in the permanent removals methodology. 2. Further assess the appropriate cut-off point in the A1 stage between carbon farming and carbon storage in products. 3. Explore methods for quantifying emissions in the A4 and A5 stages consistently and comparably, especially if not referenced from Environmental Product Declarations (EPDs). 4. Develop a framework for allocating emissions associated with building construction (A5 stage) to biobased construction elements.

3.3 Quantification of CR_{baseline}

3.3.1 Quantification of standardised CR_{baseline}

Ongoing discussion

Definition	The baselines shall be highly representative of the standard performance of comparable practices and processes in similar social, economic, environmental, technological and regulatory circumstances and take into account the geographical context including regulatory conditions.
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	The standardised baselines shall be established by the Commission in the certification methodologies [...]. The Commission shall review at least every five years and update, as appropriate, the standardised baselines in light of evolving regulatory circumstances and of the latest available scientific evidence. The updated standardised baselines shall apply only to activities for which the activity period starts after the entry into force of the applicable certification methodology.
Issue	A baseline, ‘<i>representative of the standard performance of comparable practices</i>’, needs to be clearly defined. Decisions must be made regarding the identification of these comparable activities and the functionality they are meant to fulfil. This can be interpreted as either “storage of biogenic carbon in buildings” or “storage of biogenic carbon in an equivalent biobased material, with a lifespan according to a general product pool”. Based on the interpretation of the baseline, a method needs to be specified for the operator to calculate or determine a standardised CR_{baseline}.
Objectives	An effective method must be established for calculating baseline values that are representative and adaptable to the diverse regions across the EU. These values should be determined and applied by the operator in a relevant, accurate, complete, consistent, comparable, and transparent manner.
Existing, proven certification methodologies	BBCA - Label Bas Carbone method for new buildings: Reference values for carbon stored in buildings (in kg CO₂/m²) are provided for the years 2015 and 2035, across three different building types. The 2035 values are derived through scenario development based on the official national climate strategy, allowing for linear interpolation to calculate reference values for any given year within this range. The baseline can be updated by the authority is needed. SNK – Method for carbon storage in hemp: Reference products and an average baseline value (kg CO₂/m²) are specified, calculated based on material market share. Baseline accuracy is verified every 3 years, without region-specific baselines. ONCRA - Construction Stored Carbon (CSC) concept version for 2024: The operator calculates a project-specific baseline, summing up product-level baselines of all biobased elements. Product-level baselines reflect stored carbon after 100 years, following the IPCC half-lives approach, representing materials' stored carbon in an average scenario where they are not used in the building. Timber Finance Initiative: Similar to CSC, however the difference is that the baseline doesn't represent the average carbon stored in a typical building. Instead, it reflects the average carbon stored in Harvested Wood Products (HWP) in a scenario where the HWP is not utilised as timber in structural elements but in short-lived products. The methodology defines credible data sources to determine country or region-specific decay rates for HWPs. Timber sourced from vulnerable or fire-prone forests due to climate change has a carbon removal baseline of 0. Anrechnung der Senkenleistung von Schweizer Holz als CO₂-Kompensationsmassnahme: This methodology extends beyond timber-based

	construction materials to cover all harvested wood products (wood beams, paper and furniture). Applicants can certify the carbon storage in wood products that are additional to the expected market amount. The baseline is determined from the annual volume of harvested wood products in Switzerland, based on historical official data and market trends. The market situation is reassessed annually. In the event of major changes, the baseline is reviewed by an independent panel of experts and adjusted if necessary.	
Options	Pros	Cons
The method provides reference values for stored carbon in buildings, the baseline storage factor in kg CO₂/m², differentiated across multiple building types and geographic locations. (Please note that these values are aggregated to the building level.)	• A straightforward, scalable value results in the lowest possible administrative burden for operators and auditors. • Creates a consistent and comparable calculation for all operators, irrespective of project specifics. • A methodology currently developed by Ramboll, KU Leuven, and others for DG GROW could potentially be used to calculate reference values (Ramboll, 2023). • Holds the potential to be integrated into building planning processes.	• Relatively high administrative burden on the EU: Requires data collection on material specifics and amounts in multiple regions and across multiple building types. • As market share of biobased materials evolve due to the market or regulations, the reference value can periodically be updated.
The baseline serves as a benchmark, indicating the amount of carbon that would be stored in a product pool over a specified time period if the biobased material were not utilized in the targeted building. This quantity is computed using either UNFCCC half-life rates or predetermined decay rates per material and region.	• Low administrative burden for all parties involved. • Creates consistent and comparable calculation method for all operators. • Flexible approach suitable for various construction elements. • Relatively accurate baseline can be determined for the specific project that will be certified. • Coherent approach with national inventories. • Incentivises redirection of HWP to long-lived products.	• Not in line with CRCF regulation, which requires a baseline of carbon stored in buildings, therefore cannot be used. • Accuracy depends on how well the set time period matches the lifespan of the building. • Limited material consideration in inventories: Currently, only HWP is considered in national GHG inventories for the UNFCCC; this should be expanded to include more biobased materials and be differentiated across various regions. • For additionality, it must be substantiated that the biobased material used in the building is a redirection of that material used elsewhere.
What are the options to update the baseline over time?	Pros	Cons

Revise inventory and recalculate baseline values at defined intervals (every x number of years).	• Most specific and reliable	• Highest administrative burden for EU. • Demands effective data management capabilities. • Creates uncertainties about the total amount of carbon storage that can be certified for the operator.
Establish a present baseline, project a future baseline, and interpolate linearly between the two.	• Simplicity in implementation and maintenance	• Less specific and reliable, as it heavily relies on assumptions.
Technical recommendations	1. It is recommended to define the baseline as the "carbon stored in average new buildings in a region or country", which provides a solid foundation for calculating carbon storage potential. Baseline storage factors, expressed in kg CO₂/m², need to be determined. Operators will use these factors to compute CR_{baseline}, relying on the gross floor area of their projects. Experts recommend utilising accessible tools and basing the figures on transparent studies to establish a strong foundation. 2. The market share of biobased construction elements should be periodically re-evaluated, as mandated by the CRCF Regulation, and the baseline figures for the standardised baseline adjusted accordingly. In the beginning, using projections of an evolving baseline over time is not advisable. 3. Based on the consideration that some Member States are contemplating the implementation of regular inventories and reports on the material composition of new builds and renovation projects, it is recommended to support and facilitate the establishment of such mechanisms. Combining this data with an evolving calculation methodology currently under development holds significant promise in enhancing the precision and timeliness of baseline assessments over time.	
Open questions	• Is there a necessity for future projected baselines, or are periodically updated baselines adequate? If future projections are deemed necessary, what methodology should be employed to ensure their accuracy while managing administrative burdens effectively? • How should the Commission review and update standardised baselines in light of evolving regulatory circumstances and scientific evidence? What criteria should guide this process?	
Next steps	1. Assess the necessity for future projected baselines versus periodically updated baselines. If future projections are required, determine the methodology to ensure accuracy while managing administrative burdens effectively. 2. Establish criteria to guide the process review and update standardised baselines in response to evolving regulatory circumstances and scientific evidence.	

3.3.2 Quantification of an activity-specific CR_{baseline}

Ongoing discussion

The use of a standardised baseline approach is advised. However, if it cannot be used, the operator should determine an activity-specific CR_{baseline} in the same fashion as proposed for the standardised baseline. A baseline storage factor with the unit $\text{kg CO}_2/\text{m}^2$ must be determined and multiplied with the gross floor area of the building. The operator must determine this factor based on public, verifiable sources. For instance, based on local governmental reports on biobased material use or market statistics of industry organisations, and substantiate why the used sources are recent, reliable, and applicable to the storage activity.

When an activity-specific baseline is used, uncertainty about data reliability can be a cause of concern. It is advised to automatically require the use of a discount factor when the activity-specific baseline is used. Refer to “Accounting for uncertainties in data and calculation” in this chapter for further deliberation on discount factors. Furthermore, the additionality of the storage activity needs to be tested. Refer to the chapter Additionality for additionality tests.

3.4 Quantification of CR_{total}

3.4.1 Best available practice

The best available practice involves using prescribed calculations according to EN15804+A2³ and EN15978⁴ standards used in environmental product declarations (EPDs) for estimating the biogenic carbon flows throughout a product's life cycle stages.

An Environmental Product Declaration (EPD) is a standardized document that provides comprehensive information about the environmental impacts of a product. EPDs are typically based on life cycle assessments and require third-party review. The results of an LCA are summarised in a standardised template that masks the competition-sensitive information but ensures consistency and comparability of results.

The EN15804+A2 standard outlines how EPDs for construction products should be reported. This standard specifies that the biogenic carbon content of a product should be declared as carbon removals in stage A1-A3 (production stage). This includes the total biogenic carbon stored in the raw material in stage A1 and the biogenic carbon release during the production process of the construction elements in stages A2 and A3. The total figure for stages A1-A3 is reported as 'GWP-biogenic' and expressed in $\text{kg CO}_2 \text{ eq}$. The stages A4-A5 are commonly not included in EPDs. Transport to the production site (A4) does not involve biogenic carbon emissions or removals. Construction at the site (A5) might involve further losses, these are captured in the calculation of $\text{GHG}_{\text{associated}}$ but do not have to be included here when CR_{total} is based on the biobased construction elements in the building at delivery.

³ Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products.

⁴ Sustainability of construction works - Assessment of environmental performance of buildings - Calculation method.

EPDs according to EN15804+A2 also need to declare a functional unit, such as m² of flooring, m³ of insulation material, or ‘1 product’ for prefabricated wall elements. Operators can combine EPD information with the construction project inventory to calculate the total amount of biogenic carbon stored for all biobased construction elements in the building using the formula:

$$CR_{total} = \sum_{i=1}^n CR_i * N_i$$

Where:

- CR_{total} is the total carbon removals of the carbon removal activity.
- n is the number of biobased construction elements in the building eligible for certification.
- CR_i is the biogenic carbon content of construction element i per functional unit, as reported for GWP-biogenic in stages A1-A3 from the EPD.
- N_i is the number of functional units (taken from the EPD) of construction element i in the building.

While EN standards provide a robust framework for quantifying and including biogenic carbon fluxes in the life cycle, there are open questions requiring clarification on sub-system boundaries for stages A1 to A3 and certificate awarding (refer to the ‘Quantification of GHG_{associated}’ section). Additionally, harmonisation of the applied functional unit may be necessary to ensure standardised results.

3.4.2 Potential data sources

The creation of EPDs for construction elements has become increasingly common, though it is not yet obligatory. The regulation on harmonised conditions for the marketing of construction products (CPR) will require EPD results to be available for a limited selection of products from 2025 onwards, and eventually making it mandatory for all products covered by harmonised standards. Therefore, not all eligible construction products available on the European market might have an EPD according to the standard when the certification methodology is introduced.

To allow operators to quantify the carbon storage in all eligible elements of their projects, some flexibility is required in selecting sources to determine CR_{total} . However, to maintain credibility, the methodology should offer guidance on the types of data sources, their associated data quality and how they can be used and prioritised. Figure 1 shows an example of this practice in the LCBI Certification scheme for new construction.

Priority of use	Type of data	Definition and case of use
1	Product specific EPD	Product specific EPD. The product can be manufactured at different factories but comes from one supplier. Use authorized if the commercial reference of the implemented product is covered by the specific EPD.
2	Average EPD	Represents the average of multiple products from one or more companies provided by industrial associations that cover the product. Use authorized if the commercial reference of the implemented product is covered by the list of commercial references of the average EPD.
3	Generic dataset	Environmental data in accordance with EN 15804 calculated as an average of all manufacturers of this product, representative for a country or region. This type of product data, if available, can always be used, but for more assessment accuracy, it is recommended to use a specific or average EPD when available.
4	Equivalent product EPD	This is a product specific EPD or an average EPD for an equivalent product, with the same technical characteristics and the right geographical scope of application, but it does not cover the exact product (e.g. a different brand).
5	Product specific EPD or Average EPD (other country)	A specific company product or an average EPD for a group of products from one or more companies, based on data from foreign industrial associations. Meaning the application's geographical scope of this EPD does not cover the country or region of the project. These types of data found on foreign databases can be implemented if a lack of data for the product exists on the local database. This data will be relocated by adjusting the hypothesis to the project country, including production, transport, and end-of-life scenarios.
6	Generic dataset (other country)	Environmental data in accordance with EN 15804 calculated as an average of all manufacturers of this product, based on the average of all manufacturers of this product, from another country or region that is transferred to the project's country.

Figure 1: Example of data source prioritisation from LCBI Certification scheme for new construction.

3.5 Quantification of GHG_{associated}

Ongoing discussion

Definition	GHG _{associated} is the increase in direct and indirect greenhouse gas emissions, over the entire lifecycle of the activity which are due to its implementation, including indirect land use change, calculated, where applicable, in accordance with protocols set forth in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and any further refinement.
Issue	To calculate an increase in GHG emissions could infer that an additional GHG baseline needs to be established (e.g. an increase relative to what?). The challenge then arises for a building made from biogenic material or for the individual biogenic construction element since such a baseline depends on the calculation scope (see previous section). In other words, determining which life cycle stages, according to standards and norms, should be included in this calculation becomes a critical consideration. The methodology must clearly outline how these baseline emissions should be derived to effectively if there is a GHG _{associated} .

	Further complications arise in practice, as it is possible that the $\text{GHG}_{\text{associated}}$ may be negative, especially when biobased alternatives replace concrete structural elements. Therefore, the occurrence of a ‘substitution effect’, wherein a negative $\text{GHG}_{\text{associated}}$ falsely inflates the net carbon removal benefit, needs to be addressed and prevented.	
Objectives	A method must be defined for operators to calculate $\text{GHG}_{\text{associated}}$ from baseline GHG emissions values in a relevant, accurate, complete, consistent, comparable, and transparent manner.	
Existing, proven certification methodologies	Timber Finance Initiative: Provides country-specific emission factors for common structural elements/materials, subject to periodic revisions. Riverse - Bio-based construction materials: Requires a comparative LCA based on EPDs to calculate increased emissions. The operator selects the baseline scenario and justifies its appropriateness for the case. SNK - Method for determining CO₂ emission reduction and storage in hemp: Provides default emission factors ($\text{kg CO}_2\text{eq/m}^2$) for conventional insulation materials and for the construction of two types of walls using hemp. Benchmark emissions are calculated based on the estimated reduction in the use of conventional materials for similar insulation factors. Aureus Earth - Mass Timber Building Protocol: Calculates carbon <u>avoidance</u> using default emission factors for concrete mixes ($\text{kg CO}_2\text{eq/m}^3$), differentiated over strength ranges and US regions, and steel types ($\text{kgCO}_2\text{eq/kg}$). The avoided emissions are calculated by estimating the avoided concrete use. Other reviewed methodologies do either not include the associated emissions in the quantification or include the total of all associated emissions (within scope), rather than specifically focusing on the increase.	
Options	Pros	Cons
The method prescribes a standardised $\text{GHG}_{\text{baseline}}$ expressed in $\text{kg CO}_2 \text{ eq./m}^2$ and differentiated per region and building type, similar to the baseline values given for $\text{CR}_{\text{baseline}}$. These $\text{GHG}_{\text{baseline}}$ figures must be determined for the defined scope of the calculations. Minimal the stages A1-A3 and preferable A1-A5. The applicant calculates $\text{GHG}_{\text{total}}$ emissions for the target building within the defined scope based on EPD data. For this, the sum of the indicators GWP-fossil and GWP-luluc in the stages A1-A5 for all construction elements is taken. The $\text{GHG}_{\text{baseline}}$ multiplied by the floor	- Consistent use of a standardised $\text{GHG}_{\text{baseline}}$ in combination with a $\text{CR}_{\text{baseline}}$. - Generates a lot of data on material composition of buildings.	- High administrative burden for operator to determine associated emissions for all constructions element, not only the biobased, based on EPDs. - High administrative burden on the EU and Member States to determine (and update) a $\text{GHG}_{\text{baseline}}$. - Not accurate when relying on an average value for emissions per m^2.

surface area is subtracted from the GHG_{total} to compute $GHG_{associated}$.		
The methodology prescribes default values for both carbon storage and GHG emissions for a specified list of materials and common structural elements, differentiated per region. The operator calculates $GHG_{associated}$, CR_{total} and $CR_{baseline}$ by summing all biobased structural elements. The difference with the previous option is that baseline figures per material are given instead of per building type.	- Consistent and transparent: One standardised solution for calculating $GHG_{associated}$, CR_{total}, and $CR_{baseline}$. - Low administrative burden for operators.	- High administrative burden for EU and Member States to generate and update default values for materials and structural elements across different regions. - Limits the standardised approach to the listed structural elements. - This would require the operator to define which and how many 'conventional' structural elements are replaced by biobased elements, to arrive at a baseline. This is an unrealistic representation of how buildings are designed.
The operator selects own reference scenario for biobased construction elements and calculates $GHG_{associated}$ as the 'delta' in a comparative LCA (within the prescribed system boundaries).	- Improved accuracy when the operators can choose the most likely market alternative for the construction elements to compare with. - Requires no additional information on the building's composition. - No additional administrative burden on EU. - Can be aligned with international standards for comparative LCA reporting.	- Leaves room for the operators to create unrealistic reference scenarios, resulting in less consistency and comparability among operators. - Increased administrative burden for the operator to create a comparative LCA for each biobased construction element. - Increased administrative burden for auditors to review and scrutinise all comparative LCAs.
The $GHG_{associated}$ is determined by calculating the total GHG emissions within scope. This can be calculated as sum of all GWP values in scope obtained from environmental product declarations (EPDs) of all biobased structural construction elements, utilising EPDs in accordance with EN15804.	- Calculation aligns with international standards. - Relatively low administrative burden for all parties. - Reduces discussions on baseline calculations, fostering trust with more conservative estimates. - Encourages local forestry, short chains, and efficient production by including GWP, which is deducted from storage.	This option is not in line with the regulation, as no 'increase' with respect to a baseline is calculated.
Technical conclusions	- This aspect requires further discussion. The initial option presented aligns with the recommended approach for determining $CR_{baseline}$ and meets regulatory requirements. However, it places a significant administrative burden on all parties involved. Conversely, the final option outlined in the table offers a feasible approach but does not adhere to the Regulation. - The proposed methodology should establish a standardised $GHG_{baseline}$ expressed in $kg\ CO_2\ eq./m^2$, and differentiated per region and building type. These $GHG_{baseline}$ figures must be determined within the defined scope of the calculations, encompassing stages A1-A3 at minimum and preferably extending to A1-A5. Applicants should calculate GHG_{total} for the target building by summing the indicators GWP-fossil and GWP-luluc from EPDs	

	for all construction elements in the building across stages A1-A5. $GHG_{associated}$ is then computed by subtracting $GHG_{baseline}$ multiplied by the floor surface area from GHG_{total}. Further deliberations on this matter are discussed under the section "Quantification of CR_{total}" in this chapter. 3. It is essential to establish the priority of alternative data sources in case EPDs according to EN15804+A2 are unavailable.
Open questions	• What methodologies can be employed to incorporate GHG emissions associated with indirect land use change (ILUC) into the calculation process? • Is it feasible to adopt standard emission factors for ILUC per material type, similar to those outlined for certain biofuel crops in Annex VIII of Directive (EU) 2018/2001?
Next steps	1. Further explore options for calculation of $GHG_{associated}$ to lower administrative burden for all stakeholders involved. 2. Define the priority of alternative data sources in case EPDs according to EN15804+A2 are unavailable. 3. Explore methodologies to incorporate GHG emissions associated with indirect land use change (ILUC) into the calculation process.

3.6 Accounting for uncertainties in data and calculation

Ongoing discussion

Definition	The quantification of temporary carbon storage in shall account for uncertainties in a conservative manner and in accordance with recognised statistical approaches. Uncertainties in the quantification of carbon removals shall be duly reported.
Issue	Uncertainties linked to data availability and quality pose challenges in calculating removals and embodied emissions. For insights into liability for uncertainties related to the long-term storage and risk of reversal, refer to the 'Liability' section in the Long-term Storage chapter.
Objectives	As per the CRCF regulation, these uncertainties must be addressed using established and recognised statistical approaches.
Existing, proven certification methodologies	ONCRA- Construction Stored Carbon: Uncertainties are disclosed, and in cases where precise data is unavailable, conservative assumptions are employed in calculations. Anrechnung der Senkenleistung von Schweizer Holz als CO₂-Kompensationsmassnahme: The applicant must demonstrate that the calculation method and assumptions adopted do not result in overestimation of emission reductions. Where there is a lack of precision, the applicant must implement scientific support measures.

	Riverse - Bio-based construction materials: Riverse staff reviews all applications and may request the operator to allocate 5-15% of the credits in a 'buffer pool'⁵ to account for uncertainties. Low Carbon Building Initiative (LCBI) – by Association BBCA: LCBI uses a ratings system with stars (1-7) that are granted based on data completeness. Projects with too few stars (<4) are ineligible for certification. Partially incomplete data results in the addition of “lump sums” of assumed emissions to the Embodied Carbon calculations (a variable equivalent to GHG_{associated}). The certificate is rated based on overall carbon performance (embodied, operational, stored), with a higher rating assumed to be more valuable. Royal Institution of Chartered Surveyors (RICS) - Whole life carbon assessment for the built environment: Three types of data uncertainty are distinguished; each are assigned a factor in percentages (1-15%). “Contingency factor” evaluates the design's representativeness for the final result. “Carbon data uncertainty factor” assesses the emissions data's representativeness for the actual products. “Quantities data uncertainty” gauges how thoroughly estimated or measured the reported material quantities are. The three percentages are combined to create the WLCA uncertainty factor, added to the total amount as it calculates embodied carbon, not stored carbon.	
Options	Pros	Cons
A predetermined, fixed percentage of the total carbon storage must be deducted from the certifiable amount.	Ensures a uniform approach, unaffected by variations in the operator's or auditor's assessment of data uncertainty.	Not a recognised statistical approach as intended by the CRCF regulation; functioning merely as a predetermined safety margin.
A variable percentage of the total carbon storage needs to be deducted to account for uncertainties. This percentage is based on assessment of uncertainty by the operator, guided by prescribed guidelines or assessment matrices.	- Rewards efforts to gather the most credible data. - Transparent towards operators on how reliable data gathering efforts are rewarded.	Depends on operator's own judgement posing the potential for misuse.
The certification auditor determines a percentage of the total carbon storage to be deducted. The method prescribes guidelines or assessment matrixes to determine this percentage.	Rewards efforts to gather the most credible data.	Depends on auditors' judgement, with the potential for misuse when auditors distinguish themselves from the competition based on leniency.

⁵ A buffer pool is a certified amount of stored carbon kept aside as ‘insurance’ for the certification scheme. If storage in one project turns out to be lower than expected, the buffer pool can be used to indemnify creditors and maintain the whole certification scheme’s credibility. While a discount is an amount deducted from the estimated amount of stored carbon in a project to reach a conservative estimated amount that can be certified. We propose to use deductions to account for uncertainty in the data and calculation, and a buffer pool to address liability for uncertainty over long-term storage.

Technical conclusions	1. The practice of deducting from the certifiable amount of stored carbon serves as a prevalent method for addressing uncertainties in the data.
Open questions	What criteria should be considered in determining the magnitude of deductions from the certifiable amount of stored carbon?
Next steps	1. Engage in further dialogue with operators and auditors experienced in existing methodologies to weigh the advantages and disadvantages comprehensively.

3.7 Inclusion of both new buildings and renovations

Best available practice: Include both.

Experts emphasise the substantial potential for biogenic carbon storage in renovation and retrofit projects, especially when biobased insulation materials are eligible for certification alongside structural elements (Ramboll, 2023). Therefore, it is recommended that both new buildings and renovation projects can apply for certification, provided that the most appropriate scope and calculation methods are implemented for $\text{GHG}_{\text{associated}}$, $\text{CR}_{\text{baseline}}$, and CR_{total} . Allowing certification of both new buildings and renovation projects does not require many changes to be made to the methodology.

- If the scope for $\text{GHG}_{\text{associated}}$ is set to the production and construction stages (A1-A5) of construction elements, this variable can be calculated in the same way for both construction elements in renovation projects and for new buildings.
- If CR_{total} is calculated based on the value of GWP-biogenic from EPDs, representative databases, or similar sources this variable can be calculated for both renovation and new build projects.
- For the calculation method of $\text{CR}_{\text{baseline}}$ a comparable building can be used as the reference system, a method employed by the Association BBKA. In this scenario, reference values for renovated and retrofitted buildings need to be established, potentially drawing upon the work of Ramboll on European baseline figures, which includes renovated buildings.

3.8 Cross-cutting open questions

A cross-cutting issue across all certification schemes (e.g. carbon farming, forestry and storage in biogenic building materials) is the issues relating to double counting or failure to include relevant $\text{GHG}_{\text{associated}}$ emissions, ultimately impacting the net removal values.

Throughout the discussions in the expert roundtables, the issue of double counting was a recurring concern, presenting challenges in two distinct levels: (1) within the production chain, there is the potential for double counting certified net removals due to unclear system boundaries, (2) at national level, in relation to national inventory reporting (NIR), in the case of stored biogenic carbon, this relates to the risk of overestimation of the potential harvested wood products (HWP) pool for national GHG inventories (double counting).

4. Additionality

4.1 Provisional agreement on CRCF Regulation

Article 5
Additionality

- 1. Any activity shall be additional. To that end, it shall meet both of the following criteria:
 - (a) it goes beyond Union and national statutory requirements at the level of an individual operator;
 - (b) the incentive effect of the certification is needed for the activity to become financially viable.
- 2. Where the standardised baseline [...] is used, additionality as referred to in paragraph 1 is considered to be complied with. Where the activity-specific baseline is used, additionality [...] shall be demonstrated through specific additionality tests [...] *[to be set out in the relevant EU certification methodology]*.

Aspect	Status
How can statutory additionality be safeguarded?	Ongoing discussion
How can financial additionality be safeguarded?	Ongoing discussion

4.2 Safeguard additionality

Ongoing discussion

Definition	A carbon removal activity must be "additional", meaning that it must go beyond both Union and national legal requirements and must be driven by the incentive provided by the certification. When a certification methodology uses a standardised baseline and the project performs better than the standardised baseline, this additionality is considered to be met. If, however, it is decided that an activity-specific baseline shall be used for this certification scheme, additionality must be proven.
Issue	The certification must actively encourage biogenic carbon storage in biobased construction products. It is crucial to deter operators from certifying carbon storage in their projects (and possibly monetise credits) if they would have used the biobased materials in their projects regardless.
Objectives	The concept of additionality of the carbon storage must be integral to the eligibility rules of the certification. Clear guidelines should be established to assist operators in substantiating and proving the additionality.
Existing, proven certification methodologies	ONCRA- Construction Stored Carbon: Statutorily required or (previously) subsidised carbon removals are excluded from the quantification of CR _{total} . A product class of biobased materials or construction elements with a market penetration exceeding 20% is also excluded from the quantification of CR _{total} . Operators must ensure that all biobased materials that are accounted for are not certified under any other carbon removal certification scheme.

Puro.earth Biobased Construction Materials Methodology: All load bearing engineered wooden elements are considered to be additional, provided their market penetration is below 5%. Puro.earth Woody biomass burial methodology: Operators must demonstrate financial additionality by providing full project financials and confirming that the project is not mandated by existing laws, regulations, or other binding obligations. Riverse - Bio-based construction materials: The operator must describe current or expected regulations or incentives that promote the project's use of biobased materials and prove that the use is not compelled by regulation. Furthermore, the project must prove its financial additionality by demonstrating either 1) its lack of profitability, 2) operational financial loss, or 3) the need for additional funding to facilitate short-term expansion. Alternatively, the operator must clarify why the storage activity is not common practice or describe a technological barrier and explain how carbon credit financing would enable the project to overcome this obstacle. BBCA - Label Bas Carbone method for new buildings: The operator must review and declare that local regulations do not include mandatory measures and funding relating to carbon storage. This declaration is essential to demonstrate regulatory and financial additionality. Annex VIII B of Regulation (EU) 2022/996 on rules to verify sustainability and greenhouse gas emissions saving criteria and low indirect land-use change-risk criteria: Includes a financial attractiveness test that determines whether measures to increase yield in biomass are financially attractive and would have happened without financial incentive of the certification. However, this is specific for biomass production and the added value is determined by subtracting the cost of the additionality measure from the expected income from additional biomass over the investment's lifetime, discounted at a specified rate. A building that contains biobased construction elements is not expected to generate additional income or increase in value.		
The following options are not mutually exclusive, a selection must be made to cover the various forms of additionality.		
Options	Pros	Cons
The operator must describe applicable regulations and explain how the carbon storage is additional.	- Given the absence of current regulations mandating or incentivising the use of biobased materials for this purpose, this imposes minimal administrative burden on the operator. - Member States can be asked to provide a list of local regulation and financial instruments to promote use of biobased construction elements. This can be	The process of describing and aligning with regulations might be intricate and time-consuming for operators.

	used by operator in the application process and by auditors in the verification process.	
The operator must provide project financials and a cost calculation for conventional alternatives to prove financial additionality.	Low additional administrative burden: Project developers are likely to have cost calculations readily available.	- Increased burden for the auditor to understand and review the complete set of project financials. - Operators may be reluctant to disclose sensitive financial information.
The operator must explain why the use of the biobased material is not common practice.	Can be a straightforward explanation when substantiated with market statistics or proof that a the biobased material is applied in a novel way in the building.	- Arbitrary approach that depends on arguments of operator and judgement of auditor. - Does not proof that the material choice is incentivised by the certification. - Verifying claims about common practices might be challenging.
Market penetration below a specified percentage is considered evidence of additionality.	- Low additional administrative burden on project developer and auditor. - Provides a dynamic and market-oriented approach.	- Does not prove that the material choice is incentivised by the certification. - May not capture immediate shifts in market dynamics, potentially resulting in delayed recognition of additionality. - Market penetration alone may not capture the nuances of individual projects.
The mandatory regulatory use of biobased materials and material usage financed through another certification scheme or subsidy must be declared and will be excluded from the calculation of CR_{total} .	Aligns with the challenge of deterring certification for activities that would occur regardless, ensuring regulatory compliance.	
Technical conclusions	- Given the requirement for additionality checks in certification schemes utilising activity-specific baselines, it is recommended to adopt a standardised baseline for the CRCF regulation. This ensures consistency, comparability, and transparency in calculations. - Both financial and regulatory additionality need to be covered. If Member States contribute to an exhaustive and searchable list of local regulations and financial instruments that incentivise biobased construction, the requirement for the operator to declare applicable regulation is administratively manageable. Excluding regulatory required or incentivised use of biobased materials from the calculation of CR_{total} is feasible too. - To demonstrate financial additionality, providing project financials and cost calculations for alternatives is deemed the most effective approach. However, it's acknowledged that this can be administratively burdensome and may not offer foolproof prevention against exploitation.	
Open questions	Can additionality checks be incorporated in the certification methodology, despite the stipulation in the CRCF regulation that additionality is considered to be complied with when a standardised baseline is used?	

Next steps	1. Further explore options to prove and verify financial additionality.
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5. Storage, monitoring and liability

5.1 Provisional agreement on CRCF Regulation

Article 6

Storage, monitoring and liability

1. An operator or group of operators shall demonstrate that an activity stores the carbon permanently or aims to store the carbon over the long-term.
2. For the purposes of paragraph 1, an operator or group of operators shall comply with both of the following criteria:
 - (a) they shall be subject to rules to monitor and mitigate any identified risks of reversal occurring during the monitoring period;
 - (b) they shall be liable to address any reversal of the carbon captured and stored by an activity, occurring during the monitoring period, through appropriate liability mechanisms as set out in the delegated acts adopted pursuant to Article 8.
- (2a) The monitoring rules referred to in paragraph 2, point (a), shall:
 - (c) for carbon farming and carbon storage in long lasting products, be set out in accordance with the rules laid down in the certification methodologies set out in the delegated acts adopted pursuant to Article 8.
- (2b) The liability mechanisms referred to in paragraph 2, point (b), shall:
 - (c) for carbon storage in long lasting products and for carbon farming, be set out and duly justified in the applicable certification methodology and may include up-front insurance or collective buffers.

The conditions outlined in Article 6(1) can be translated into eligibility requirements, specifying that only biogenic carbon storage within structural construction elements and insulation materials can be certified. It is determined that eligible construction elements must have a minimal expected lifespan of 35 years.

Aspect	Status
Monitoring protocol	Ongoing discussion
Liability for unplanned release of carbon	Ongoing discussion
Inclusion of risk mitigation in certification	Open question
Partial rebuilding or refurbishment during certification period	Best available practice
Incentive to prolong storage time	Open question

5.2 Monitoring protocol

Ongoing discussion

For clarification purpose – this phase of the certification process refers to the operational phase of the building (i.e. B1-B5).

Definition	An operator or group of operators shall monitor and mitigate any risk of release of the stored carbon occurring during the monitoring period.	
Issue	During the certification period and at the moment of recertification, the target building must be monitored to verify the sustained presence and functionality of the certified amount of carbon or the certified biobased elements.	
Objectives	A monitoring protocol needs to be developed, specifying the interval at which it should be checked whether the certified carbon remains stored within the building.	
Existing, proven certification methodologies	Timber Finance Initiative: The lifespan of the certified element is assumed to be equivalent to the building's lifespan, which is verified using satellite photographs. ONCRA- Construction Stored Carbon: The monitoring period is 100 years irrespective of the (expected) lifespan of the project, with monitoring conducted once a year. Riverse - Bio-based construction materials: The monitoring period is equal to the certification period. The certification body determines a case-specific monitoring frequency ranging from 3 months to 1 year.	
Options	Pros	Cons
The assumed storage time for the carbon aligns with the lifespan of the building. This can be effectively monitored by requiring the operator to demonstrate the continued existence of the building, potentially through means such as satellite photographs.	Low administrative burden and cost-effective, reducing barriers to certification.	- Inaccurate approach for structural elements that might be replaced during the building's lifespan. - The frequency for monitoring still needs to be determined.
A predefined monitoring frequency is established for conducting inspections to ensure the ongoing carbon storage in the construction elements.	Transparent and straightforward method, ensuring consistency across various projects.	- May be excessive for projects with a very low risk of reversal, potentially resulting in unnecessary administration and costs. - Opting for a longer interval might pose the risk of unregistered reversal in the interim.
Technical conclusions	The CRCF regulation mandates that all activities undergo periodic recertification audits at least once every five years. This cadence is considered adequate for monitoring biobased construction products, particularly when eligibility is restricted to structural elements and insulation materials.	

5.3 Liability for unplanned reversal

Ongoing discussion

Definition	An operator or group of operators shall be subject to appropriate liability mechanisms in order to address any release of the stored carbon occurring during the monitoring period.	
Issue	Risks such as fire or premature demolition pose threats to the stored carbon but also to the credibility of the certification. For insights into liability for incorrect quantification of the net carbon removal benefits, refer to the 'Accounting for uncertainties' section in the Quantification chapter.	
Objectives	Liability mechanisms designed to indemnify certificate holders in case of unplanned reversal.	
Existing, proven certification methodologies	Puro.earth Biobased Construction Materials Methodology: A 'buffer pool' is integrated into the methodology to account for uncertainties about loss after production and other potential losses. The standard percentage for this buffer is set at 10%, subject to adjustment by the auditor. Verra VCS Standard: Using a risk assessment with a 'non-permanence' risk tool is required. This tool determines the number of credits to be deposited in a buffer pool (specifically for agriculture, forestry, and other land use projects). Riverse - Bio-based construction materials: A standard 10% of the credits is reserved in a 'provisions pool' to mitigate identified risks. BBCA - Label Bas Carbone method for new buildings: A 10% discount on the total stored amount (equivalent to CR_{total}) is applied to cover all risks associated with the project. The liability mechanisms listed below are similar to the approaches to deal with data uncertainty in the quantification of the net carbon removal benefit. However, unlike discounts, these methodologies exclusively utilise solutions involving a 'buffer pool.' This pool represents a certified amount of stored carbon set aside as 'insurance' for the certification scheme. In the event of unintentional reversal in one project, the buffer pool can be leveraged to indemnify creditors and maintain the overall credibility of the certification scheme. This stands in contrast to discounts, which deduct an amount from the estimated stored carbon, providing a conservative certification estimate but lacking the ability to address the problem when it occurs.	
Options	Pros	Cons
A set percentage of the total certified units must be allocated to a buffer pool.	Consistent approach, irrespective of the operator's or auditor's risk assessment.	May be considered a blunt approach, necessitating clarification regarding the basis for determining the percentage.
A varying percentage of the total certified units needs to be placed in a buffer pool. This percentage is	Transparent for operators, outlining how risks are evaluated.	Increases administrative complexity for operators.

contingent upon the operator's risk assessment. The method provides guidelines or assessment matrices to ascertain this percentage.		• Depends on the operator's own judgement, posing the potential for misuse.
The certification auditor allocates a percentage of the total certified units to a buffer pool, guided by prescribed methods and assessment matrices.	• Combines project-specific risks with independent judgement.	• Increases administrative complexity for both the operator and auditor.
Technical conclusions	1. After reviewing existing methodologies, no specific liability mechanisms were found, except for buffer pools used as 'insurance' against unplanned releases. Experts hesitated on the suggested buffer pool of 5-10%, fearing it could reduce certification incentives and hinder biobased product use. To mitigate this issue, a proposed solution involves leveraging national insurance data to assess the region-specific risk of reversal. By incorporating such data, a more tailored and nuanced approach to establishing buffer pools can be achieved. This ensures risk mitigation without dampening certification motivation.	
Open questions	• What factors should be taken into account when determining the ideal percentage or range of percentages for a buffer pool? • It needs to strike a balance between adequately covering potential losses from unplanned releases across the certification program while also maximising incentives for certification and the adoption of biobased construction materials. How can we ensure that this balance is achieved effectively?	
Next steps	1. Alternative options for liability mechanisms must be explored in a next stage of the assessment. One proposed option is to investigate the feasibility of establishing insurances through insurance agencies to cover the financial liability for unplanned reversal of carbon storage. This entails conducting further research into the practicality, cost-effectiveness, and potential benefits of this approach to determine its viability for implementation.	

5.4 Risk mitigation

Open question

Unplanned release risks, such as fire or premature demolition, pose threats to the long-term stored carbon. Mitigation and prevention plans must be in place to reduce the risk and impact of such unplanned releases from a completed building project. Two examples from existing certification schemes were found. Both use a list of potential risks that need to be considered by the operator (explained below). The liability for unplanned reversal has been discussed however, an open question remains about which risk mitigation mechanisms are best to use and therefore, this requires further deliberation.

The Puro.earth Woody biomass burial methodology addresses risks within the certification eligibility rules by employing a risk and mitigation matrix. Some identified risks lead to eligibility requirements to the projects for certification. Others need to be considered in a 'risk mitigation plan' that needs to be delivered when applying for certification. The identified risks and mitigation strategies, initially designed for biomass burial, are partially applicable to carbon storage in construction products.

The Riverse methodology bio-based construction materials requires an evaluation of the risk of reversal using a prescribed format that outlines potential causes and their likelihood. The operator has to indicate the level of risk for each potential threat with a number, scaled from 0 (no risk at all) to 10 (high risk). In the certification process the operator is required to deliver a risk assessment that includes the results.

5.5 Partial rebuilding or refurbishment during certification period

Best available practice

Concerning monitoring and liability, experts have raised questions regarding how the refurbishment or partial rebuilding of a construction project during the certification period would be addressed. The expert feedback indicated that these activities should not fall within the scope of GHG_{associated}, as outlined in the Quantification chapter. This decision stems from the inherent difficulty in accurately predicting and quantifying the repair, refurbishment, and replacement of parts over the building's lifespan.

The concerns raised by the experts were not about accurate quantification but about the issuance of certificates for theoretically stored carbon that might be released in practice due to replacements during the certification period. To mitigate this risk, the use of time-limited certificates and a restriction to structural elements is incorporated in the CRCF regulation. To further safeguard against this risk, a provision can be included, stipulating that any removal of biobased elements included in the certification would automatically trigger certificate withdrawal, necessitating recertification.

5.6 Incentive to prolong storage time

Open question

Further discussion is needed on how to incentivize building owners through certification to extend the storage of carbon in a building beyond the minimum requirement of 35 years. The aim of the certification is to encourage and facilitate the long-term storage of biogenic carbon in construction products. The current proposal suggests certifying biobased construction products with a minimum lifespan of 35 years and implementing time-limited certification. Re-certification audits will occur at least every 5 years. This strategy ensures a climate benefit over the 35-year storage period and includes monitoring as frequent as the certification period, providing assurance of the stored carbon during re-certification.

However, under this approach, project developers or certificate holders do not gain any advantage from a storage time exceeding the 35-year minimum. This has led to the exploration of how the certification mechanism can incentivize even longer storage times.

One potential solution is to gradually release certificates from a buffer pool over time after the initial 35 years, rewarding building owners for extended lifespans. Alternatively, additional value could be assigned to certificates for projects with a lifespan exceeding 35 years through the co-benefits mechanism. Various approaches can be developed and evaluated to prevent unintended consequences.

6. Sustainability

6.1 Provisional agreement on CRCF Regulation

Article 7
Sustainability

1. An activity shall not significantly harm and may generate co-benefits for one or more of, the following sustainability objectives:
 - (a) climate change mitigation beyond the net carbon removal benefit and net soil emission reduction benefit referred to in Article 4(1) and (1a);
 - (b) climate change adaptation;
 - (c) sustainable use and protection of water and marine resources;
 - (d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials;
 - (e) pollution prevention and control;
 - (f) protection and restoration of biodiversity and ecosystems including soil health, as well as avoidance of land degradation.
2. For the purposes of paragraph 1 of this Article, an activity shall comply with minimum sustainability requirements laid down in the certification methodologies set out in the delegated acts adopted pursuant to Article 8. The minimum sustainability requirements shall take into account the impacts both within and outside the Union and local conditions. Those minimum sustainability requirements shall, where appropriate, be consistent with the technical screening criteria for the ‘do no significant harm’ principle. The minimum sustainability requirements shall promote the sustainability of forest and agriculture biomass raw material in accordance with the sustainability and GHG saving criteria for biofuels, bioliquids and biomass fuels laid down in Article 29 of Directive (EU) 2018/2001.
3. Where an operator or group of operators reports co-benefits [...] beyond the minimum sustainability requirements referred to in paragraph 2 of this Article, they shall comply with the certification methodologies set out in the delegated acts adopted pursuant to in Article 8. The certification methodologies shall include elements to incentivise as much as possible the generation of co-benefits going beyond the minimum sustainability requirements, in particular for the objective referred to in paragraph 1, point (f), of this Article.

Aspect	Status
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Minimum sustainability requirements	Ongoing discussion
Selection of co-benefits	Open question
Incentivise generation of co-benefits	Open question
Monitoring of minimal requirements and co-benefits	Open question

6.2 Minimum sustainability requirements

Ongoing discussion

The CRCF regulation requires the certification methodology to establish minimum sustainability requirements across the six environmental objectives of the EU Taxonomy. These requirements are set as thresholds that should at least be reached for a project to be eligible for certification. This is included to prevent that the incentive for carbon storage in products harms the environmental objectives of the Taxonomy. Experts have mainly expressed their concern about potential threats to biodiversity and with the incentivisation of biobased materials. This threat can for instance be mitigated with the appropriate eligibility requirements related to biodiversity and ecosystems protection and DG CLIMA has expressed the intention to align the minimum sustainability requirements (and other parts of the certification scheme) with existing EU regulation and relevant initiatives to the greatest extent possible.

While this section provides a list of some relevant EU regulations and initiatives, it is not exhaustive, and further exploration and discussion are essential for both minimum requirements and existing regulations and initiatives.

Specifically, further alignment with the ‘Do no significant harm’ (DNSH) criteria from the Taxonomy and the macro-objectives from Level(s) should be explored. The DNSH criteria mandate that an activity should do no significant harm to any of the six environmental objectives of the EU Taxonomy to be considered sustainable. Thresholds are defined that cannot be crossed per objective in technical screening criteria.

Level(s) is a voluntary assessment and reporting framework for the sustainability performance of buildings. It is based on six macro-objectives, from which four match with the environmental objectives for the minimum sustainability requirements; climate change mitigation, climate change adaptation, efficient use of water resources, and resource efficient material use. It is a voluntary framework that does not provide targets, but it does provide indicators for each objective and methodologies to assess these. The thresholds for the indicators can be determined for this certification program.

Climate change mitigation

For sustainable building use, a key requirement is meeting a specific energy performance standard outlined in the Energy Performance of Buildings Directive (EPBD; [EU/2010/31](#)) for the target building.

Climate change adaptation

While no specific regulations pertaining to buildings or construction were identified for this objective, a connection can be established with the voluntary [Technical guidance on adapting buildings to climate change](#).

[Sustainable use and protection of water and marine resources](#)

The target building must adhere to the basic requirements set out in annex 1.1 of the [Proposal](#) for harmonised conditions for the marketing of construction products (CPR). These requirements focus on preventing the release of hazardous substances to groundwater, marine, or surface waters, and minimising the overall use of drinking and brown water throughout the product's lifecycle. Development of harmonised technical specifications and standards will be essential.

[Transition to a circular economy](#)

A minimum requirement concerning the circular economy or circular design can be linked to CPR. This could be through the requirement that digital product passports are available for all construction elements that are included in the calculation of CR_{total} enabling better demolition and sorting at end of life, or compliance to the inherent product environmental requirements set out in Annex 1.3.3 of the CPR.

In the Riverse - Bio-based construction materials certification, the material footprint and tons of avoided waste per year must be provided in the certification application. However, these will not be factored into the calculation of certifiable storage.

[Pollution prevention and control](#)

Similar to the sustainable use and protection of water and marine resources, this objective can be linked to the requirements in the [Proposal](#) for harmonised conditions for the marketing of construction products. Further development of these requirements is necessary through harmonised technical specifications and standards.

[Protection and restoration of biodiversity and ecosystems](#)

The 'Sustainability and greenhouse gas emissions saving criteria for biofuels, bioliquids and biomass fuels' from the Renewable Energy Directive ([RED III](#)) can be used as a template to establish minimum requirements regarding this sustainability objective. While the direct linkage of minimal sustainability requirements to this regulation faces challenges due to the diverse uses of biomass—ranging from raw material to energy—Article 29 of RED III offers a solid foundation.

Article 29(3) of RED III prohibits the use of biomass from certain biodiverse areas for certified renewable energy, safeguarding areas recognised as highly biodiverse by competent authorities. This includes stipulations like refraining from sourcing raw material from *“biodiverse forest and other wooded land which is species-rich and not degraded, and has been identified as being highly biodiverse by the relevant competent authority, unless evidence is provided that the production of that raw material did not interfere with those nature protection purposes”*.

Article 29(6) of RED III sets out criteria to prevent the use of forest biomass from unsustainable production, emphasising practices that *“maintains or improves the long-term production capacity of the forest”*. It mandates operators to demonstrate sustainability compliance. Similarly, Article 29(7) outlines criteria for land-use, land-use change and forestry, requiring local authorities to provide *“evidence that reported LULUCF-sector emissions do not exceed removals”*.

These options would allow for the utilisation of timber and other biobased materials from outside the EU. However, some experts advocate limiting eligibility to materials sourced within the EU. Their argument centres on the ease of guaranteeing the sustainable sourcing of materials and the enhanced traceability of material flows for national GHG inventories.

Most existing certification programs lack explicit and well-defined minimum sustainability requirements. Sustainability assessments often indirectly rely on the availability of certifications like PEFC or FSC for wood products, and assuming forest regrowth in their eligibility criteria. Starting at the forest level, PEFC and FSC certify forest management units (FMU). An FMU is a designated area of forestland. With chain of custody certification, the timber originating from the FMU is tracked as it moves through manufacturing, and distribution stages. The chain of custody certification of the construction products does not identify the FMU where the timber originated from but does verify that the value adheres to social and environmental standards too. To be certified the FMU has to adhere to social and environmental criteria. The specific criteria differ per region or country, but generally require that the forestry operation biodiversity and forest cover, protects water resources, conserves soil quality, and excludes areas with high conservation value or protected biodiversity.

6.3 Reporting of co-benefits

Open question

The reporting of co-benefits for the projects is a crucial step that remains to be determined. Both the Riverse and Association BBKA certification methodologies present potential co-benefits for project consideration. While these lists serve as inspiration for choosing co-benefits related to construction projects, it's important to note that the co-benefits identified by these methodologies do not fully match the sustainability objectives outlined in the CRCF regulation. The co-benefits identified in the existing methodologies are worthy goals, but the regulation specifies that the certification methodologies shall include elements to incentivise co-benefits for the six environmental objectives of the EU Taxonomy.

The Riverse methodology lists 21 broadly defined objectives based on the SDGs. It requires project-specific translation of these objectives by the operator to verifiable co-benefits. On the other hand, the methodology by Association BBKA provides a list of co-benefits along with reporting methods, presenting a more consistent, comparable, and transparent approach across different projects and operators. This aligns better with the CRCF regulation.

During the discussions, several co-benefits were suggested for inclusion. To address the objective of climate change mitigation, projects could be scored based on the low embodied carbon of the certified construction elements or by measuring the whole-life carbon (WLC) of the project, including operational carbon. For the objective of transitioning to a circular economy, projects or included construction elements could be scored on recycled content use, recyclability, or reusability. An overarching co-benefit could be the promotion of sufficiency, emphasising the lowest resource or raw material use to fulfil the function. If sourcing raw materials from the EU is not set as an eligibility criterion, the use of locally sourced crops can also be included as a co-benefit.

6.4 Incentivise generation of co-benefits

Ongoing discussion

Definition	The certification methodologies shall incentivise as much as possible the generation of co-benefits going beyond the minimum sustainability requirements, in particular in pursuit of the objective referred to in paragraph 1, point (f) Protection and restoration of biodiversity and ecosystems.	
Issue	The certification methodology is primarily designed to incentivise long-term carbon storage in products. The driving force behind this incentive lies in the financial potential of certification within the voluntary market. Although projects are required to meet minimum sustainability standards, there is currently no added incentive in the regulation for project owners to surpass these requirements and secure additional co-benefits. Since the certification will be issued on project level, the protection and restoration of biodiversity and ecosystems is a challenging co-benefit. Minimal requirements on sustainably harvested raw materials can be set, but further protection and restoration of biodiversity and ecosystems is hard to incentivise.	
Objectives	A method must be developed for the reporting of co-benefits.	
Existing, proven certification methodologies	LCBI Label: LCBI doesn't specifically incentivise co-benefits but has a robust system promoting comprehensive reporting, minimising emissions, and impacts while maximizing the carbon storage. A project can earn a number of stars based on the completeness of data. The project is further scored on embodied carbon, operational carbon, and stored carbon, each with set thresholds. These stars and carbon performance collectively determine the certification's rating: Standard, Performance, or Excellence, adding inherent value to the certificate. Riverse - Bio-based construction materials: Each project is required to report on at least two quantifiable and verifiable environmental or social co-benefits. The selection can be made from a list of 21 options aligned with the UN Sustainable Development Goals (SDGs). For example, choices include "Improve global resource efficiency in consumption and production" or "Ensure the conservation, restoration, and sustainable use of terrestrial and inland freshwater ecosystems and their services." Operators must choose two co-benefits and devise a monitoring plan to report progress on these co-benefits. BBCA - Label Bas Carbone method for new buildings: Four categories of co-benefits are identified: Biodiversity, Environment, Adapting to climate change, and Socio-economic. A list of 19 co-benefits is provided across these categories, each with means to substantiate their inclusion. For example, one criterion is, "Construction companies working on the site are located between 50 and 100 km from the site as the crow flies." Each included co-benefit earns project points, with each category having a maximum score. Operators report earned points as a percentage of the maximum achievable, serving to showcase the environmental and social quality of a project and attracting potential financiers.	
Options	Pros	Cons

The methodology sets tiers of certification , graded according to the inclusion and generation of co-benefits. It is presumed that a 'gold status' certificate holds greater value than a 'silver status', highlighting the prominence of co-benefits in the certification system alongside carbon storage.	The mention of a particular certificate tier brings the co-benefits into sharp focus, positioning them as equally crucial as carbon storage within the certification system.	Tiers of certification based on co-benefits could potentially introduce confusion regarding the type of storage or cast doubt on the credibility of the certificate.
Projects can earn points (or stars) across predefined categories of co-benefits they generate, a feature prominently displayed on the certificate to underscore the inclusion of benefits beyond carbon storage.	Inclusion of points or stars offers a subtle acknowledgment of co-benefits without placing excessive emphasis on them.	This method might still induce confusion regarding the nature of storage or the certificate's credibility. The points or stars for co-benefits can be confused with the permanency of the storage, or reliability of the quantification.
The method requires the operator choosing one or more co-benefits that the project aims to generate and subsequently reporting on these selections. This obligation can be approached with a best-effort mindset, without a specified minimum threshold.	Mandating co-benefits ensures that every project generates some form of additional advantages while maintaining flexibility for the operator.	Requiring mandatory co-benefits adds another layer of obligation for the operator, potentially creating obstacles or distractions from the primary certification goal.
Technical conclusion	The cross-cutting nature of the topic necessitates further discussion to arrive at comprehensive conclusions.	
Open questions	How can the reporting on contributions to co-benefits be effectively mandated in the certification application process?	
Next steps	- Facilitate additional discussions among stakeholders to delve deeper into the implications and feasibility of mandating reporting on contributions to co-benefits. - Conduct a thorough analysis to identify the most relevant and impactful co-benefits to include in the reporting process. - Explore potential mechanisms for enforcing compliance with reporting requirements while minimizing administrative burden on operators.	

6.5 Monitoring of minimal sustainability requirements and co-benefits

Open question

In addition to monitoring carbon storage, it is crucial to track minimal sustainability requirements and co-benefits throughout the activity period. The CRCF regulation specifies that the carbon storage should allow for on-site monitoring. Depending on further definition, it is likely that the minimal sustainability requirements and co-benefits will likely not all allow for monitoring on-site. How can monitoring of these requirements and benefits be included in the regular monitoring process? The two existing certification methodologies mentioned earlier, which already incorporate a co-benefit mechanism, offer insights into how sustainability requirements and co-benefits can be effectively monitored.

Riverse - Bio-based construction materials

The Riverse staff determines a reporting frequency for each project, ranging from every 3 months to once a year. Operators are required to update progress on co-benefits through an online portal. This approach is suitable for sustainability objectives that evolve over the certification period, such as the gradual restoration of biodiversity in the area of the construction project, measurable over time.

Association BBKA - Label Bas Carbone method for new buildings

The indicators are documented in a monitoring report, which is subsequently submitted to an auditor. The validation of these indicators, along with the necessary supporting evidence, is carried out by an auditor during the verification of emissions reductions. The audit is performed once, initiated at the earliest upon the building's completion and concluding no later than two years post-completion. This method is particularly fitting for sustainability objectives that remain constant, such as achieving a minimum or threshold whole building embodied carbon value.

7. Glossary

Activity period	Period over which the carbon is stored in the construction elements in the building.
Avoided carbon	Avoided carbon refers to the reduction in greenhouse gas emissions achieved when low-carbon products replace more emission-intensive materials. This is commonly referred to as the substitution effect.
CRCF	Carbon Removals Certification Framework (CRCF) is a legislative initiative within the European Union that defines and formulates methodologies for certifying carbon removals.
DG GROW	Directorate-General of the EU for Internal Market, Industry, Entrepreneurship and SMEs
Embodied carbon	Embodied carbon encompasses greenhouse gas emissions resulting from the production, usage, and disposal of a construction element over its lifecycle.
EN15804+A2	European norm for Environmental Product Declarations (EPDs) core rules for the product category of construction products.
EPD	Environmental Product Declaration (EPD) is a standardised document quantifying environmental information on a product's life cycle, facilitating comparisons between products with similar functions.
GHG	Greenhouse Gas (GHG) is a collective term for gases that trap heat in the Earth's atmosphere, contributing to the greenhouse effect.
GWP	Global Warming Potential (GWP) measures the heat-trapping capability of greenhouse gases over a specified time period, usually 100 years. It is a key metric used in climate change assessments and comparisons between different gases.
GWP-biogenic	The GWP-biogenic indicator, used in Life Cycle Assessments (LCAs) and EPDs, measures CO ₂ absorbed by biomass and CO ₂ equivalent greenhouse gas emissions from stored biogenic carbon.
GWP-luluc	The GWP-luluc (Global Warming Potential from Land Use and Land Use Change) measures the greenhouse gas emissions and removals associated with changes in land use, forestry activities, and other land management practices
LCA	A life cycle assessment (LCA) is a systematic methodology that evaluates all stages of a product's life, including extraction, production, transportation, use, and disposal.

Monitoring period	Period over which the storage of carbon is monitored by an operator as determined in the certification methodology
Operator	Any legal or natural person or public entity who operates or controls the storage activity, or to whom decisive economic power over the technical functioning of the activity has been delegated. In this specific case: project developer or building owners who apply for certification.
PEF	The Product Environmental Footprint is a standardised LCA methodology developed by the EU's Joint Research Committee as a common way of measuring environmental performance of products.
Scope	"Scope" refers to the different stages of the building process defined in standard EN 15804 that the methodology aims to encompass.
SDGs	Sustainable Development Goals (SDGs) are a set of 17 goals established by the United Nations to promote peace, prosperity, and sustainability for people and the planet.
Stored carbon	In this context, stored carbon refers to the biogenic carbon sequestered by natural processes in materials, remaining locked within bio-based construction products throughout the building's lifespan.
WLC	Whole Life Carbon (WLC) represents the combined total greenhouse gas emissions of embodied and operational emissions over the entire life cycle of a building.